

Bitcoin Relaying Protocol Litepaper v1.0

BIFROST FOUNDATION

1 Introduction

Bitcoin (BTC) is both iconic and paradoxical. As the most secure and widely recognized cryptocurrency with the largest market capitalization, it continues to gain traction in traditional finance. Yet, its limited functionality restricts its broader utility. This has inspired the concept of “BTCFi,” which aims to extend Bitcoin’s financial use cases and aligns with the vision behind the services we build on the Bifrost Network.

One of the recurrent challenges in securely connecting BTC, such as via cross-chain bridges, is the risk posed by off-chain participants like relayers. While on-chain operations are inherently authenticated and verified by the blockchain network, off-chain operations often introduce vulnerabilities due to their dependence on trust and susceptibility to manipulation or failure. Therefore, ensuring transparency in off-chain operations is crucial to mitigate these risks. Many BTC-connecting protocols, such as Bitlayer, BSquared, Citrea, and Merlin, strive to enhance the transparency of their off-chain operations.

We took a more straightforward and secure approach. Our Bitcoin Relaying Protocol (BRP) performs the entire end-to-end process on-chain by design. This ensures transparency and public verifiability, effectively eliminating the risks posed by off-chain actors. On-chain operations provide a trustless environment where every step is recorded on blockchains, aiming for higher reliability and security.

BRP connects the Bitcoin and Bifrost networks by adhering to the following design principles:

- **On-chain Completeness:** Every action in the cross-chain process should be recorded on-chain, ensuring public verifiability.
- **User-accessible Transparency:** Due to the cross-chain nature, actions are inevitably recorded in different blockchains. However, we make each action to be recorded on the most informative blockchain possible.
- **Seamless Operation:** Although a cross-chain process involves on-chain operations from different blockchains, we design it to proceed without requiring additional user’s intervention once started.

Connecting the Bitcoin network to the Bifrost network is like combining two different worlds; they have different message formats and cannot communicate without intermediaries. We eliminate this intermediary off-chain gap by letting the relayers communicate and collaborate entirely through blockchains. The Bifrost network, being a dual-stack blockchain of EVM (Ethereum Virtual Machine) and Substrate, allows relayers to make an on-chain consensus on the EVM side while executing the cross-chain operations on the Substrate side.

Among the two stacks, EVM is more familiar to users, but its functionality is limited due to the restrictions of the EVM’s design. Substrate, on the other hand, supports more complex operations while managing blockchain state more directly. Following the design principles of BRP, we enable consensus operations to be validated on on-chain smart contracts on the EVM. However, handling Bitcoin transactions is too complex for the EVM. For instance, cryptographic operations require cryptographic libraries that are not supported in the EVM. Therefore, the Bitcoin components of the cross-chain process are executed via Substrate transactions.

To the best of our knowledge, BRP provides the most transparent end-to-end process for connecting BTC. In the following sections, we will explain how BRP achieves on-chain operability with Bitcoin.

2 On-chain Coverage

The off-chain relayer model transfers BTC from the Bitcoin network to other blockchains. Figure 1 illustrates the off-chain relayer model, commonly used in most cross-chain protocols, such as Bitlayer, BSquared, Citrea, and Merlin. When a user withdraws BTC from a blockchain to the Bitcoin network, the process starts by emitting a withdrawal event to notify the relayers operating off-chain. The relayers communicate with each other about

their findings, sensing the event from the origin network to reach a consensus by communicating with each other. They then prepare a transaction to transfer BTC from their BTC vault to the user's BTC address on the Bitcoin network. The BTC vault is usually a multi-signature (multi-sig) wallet but is implemented differently depending on the protocol using variants from Shamir Secret Sharing (SSS) or threshold signature schemes.

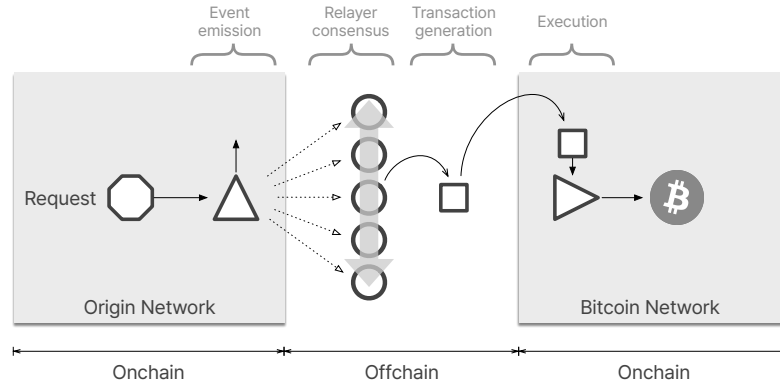


Fig. 1. Reference Off-chain Relay Model (to the Bitcoin network)

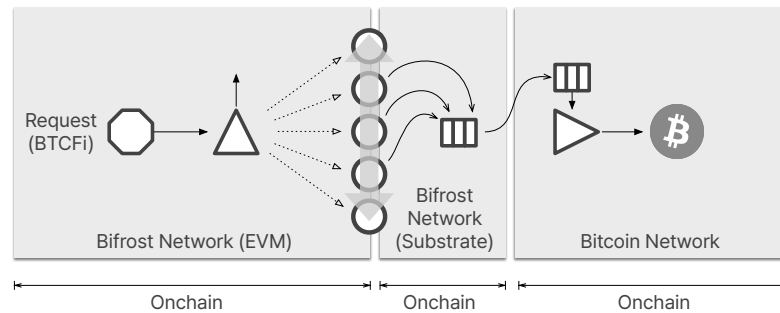


Fig. 2. BRP Model (to the Bitcoin network)

These steps by the off-chain relayers are in the shadows. Decentralized protocols rely on the assumption that the system follows the majority opinion. However, this might be misleading without transparency on the entire process. One may be concerned that only a few relayers participate in signing to control the BTC vault and that no information is provided about who is participating.

In BRP, the relayers act on-chain on both the EVM and Substrate stacks as illustrated in Figure 2. The Bifrost relayers read the BTC withdrawal event from the vault contracts and send a vote transaction to the system contract on the EVM stack of the Bifrost Network until the vote reaches a quorum, i.e., the relayers reach a consensus that the BTC withdrawal is valid. If the vote reaches a quorum, the relayers sign the BTC transfer transaction on the Substrate stack of the Bifrost Network. We choose to use Segwit multi-sig for the BTC vault for the transparency of signers. Thus, every activity of the relayers is recorded on-chain sequentially. In this outgoing scenario, the steps for the relayer consensus, the BTC transaction generation, and the BTC transfer are executed on-chain on the EVM and Substrate stacks (Bifrost Network) and the Bitcoin network.

3 System Architecture

BRP is engineered as a decentralized protocol, underpinned by the architecture of Bifrost Network, a decentralized platform facilitating cross-chain communication. Figure 3 illustrates the comprehensive system architecture of BRP. This architecture comprises various components, including Bifrost relay, components within Bifrost node, PSBT manager, Bitcoin core, and a threat monitoring system known as Watchtower. Subsequent sections will provide detailed descriptions of each component.

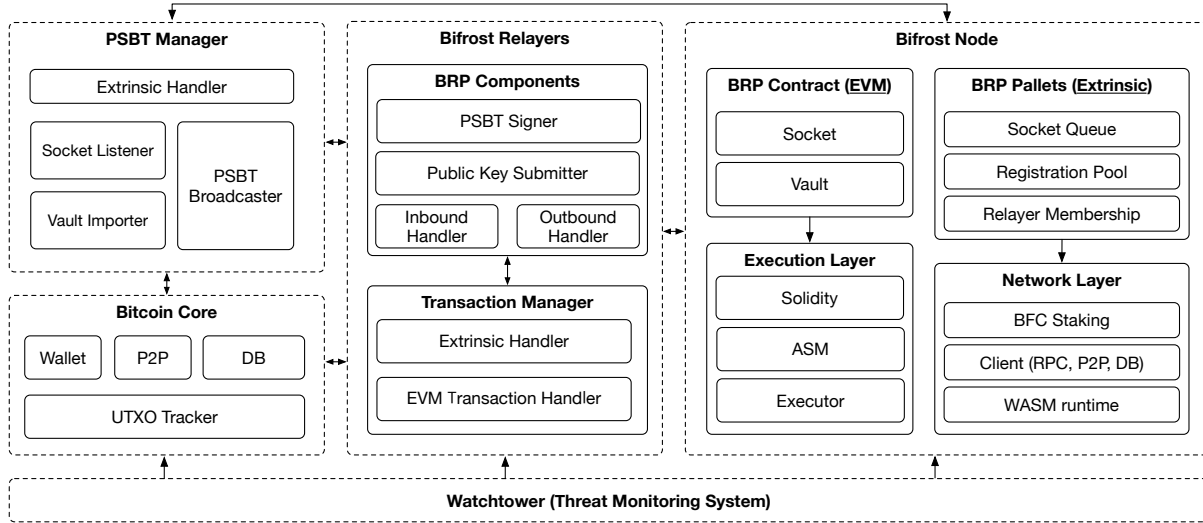


Fig. 3. Bitcoin Relaying Protocol System Architecture

3.1 Bifrost Relay

Bitcoin lacks inter-chain communication capabilities; therefore, we have implemented an off-chain relaying mechanism called relay. It is tasked with monitoring and transmitting transactions between Bitcoin and Bifrost Network. Additionally, since BRP operates atop Cross-Chain Communication Protocol (CCCP)¹, relay also facilitates inter-chain communication with other blockchains, including Ethereum, Base, Arbitrum, BSC, and Polygon.

We aimed to design the cross-chain communication protocol, such as BRP and CCCP, in a way that minimizes the responsibilities carried out by off-chain components like relay. Consequently, relayers do not verify the integrity of protocol-related event messages; instead, they solely relay BRP-related events to Bifrost Network or Bitcoin. This design choice ensures that, even if a majority of relayers are compromised and able to pause the system, they cannot manipulate the integrity of BRP-related messages. Instead, the BRP smart contract on the Bifrost Network verifies the integrity of the relayed events. For instance, consider a scenario where a user intends to bridge their BTC to Bifrost Network (inbound transaction). Upon detecting a vault deposit event from either Bitcoin or Bifrost Network, relayers encapsulate the event into BRP transactions and forward them to the BRP smart contract on Bifrost Network. Subsequently, contracts aggregate these transactions forwarded by each relay. If a BRP transaction from a relay achieves a majority consensus, BRP contracts will execute the corresponding logic,

¹<https://docs.bifrostnetwork.com/bifrost-network/bifrost-network-architecture/cross-chain-communication-protocol-cccp>

such as bridging assets from Bitcoin to Bifrost Network. The detailed design of each component is explained in the following sections.

3.1.1 PSBT Signer. PSBT, or Partially Signed Bitcoin Transaction, is a format that facilitates exchange of transaction information between parties without disclosing full details. PSBT signer is tasked with signing these transactions for Bitcoin. In the case of an inbound bridge, BRP sets up a secure deposit vault on the Bitcoin by utilizing a multi-signature system, which involves keys from relayers. If a majority— m out of n signers (relayers)—agree, BTC held in BRP Vault (a multi-signature-based vault) is transferred to the designated recipient. This mechanism safeguards against malicious actions by minorities, ensuring that locked assets remain secure even if some signers are compromised.

3.1.2 Public Key Submitter. BRP utilizes public keys of n relayers to establish a multi-signature vault on the Bitcoin. For this process, we introduce two vault generation mechanisms: 1) on-demand key submission and 2) pre-key submission. The on-demand key submission mechanism requires relayers to submit their public keys upon a vault generation request. In contrast, pre-key submission involves relayers submitting their public keys in advance. This approach ensures that vault can be generated immediately upon receiving a vault generation request by using pre-submitted keys, which accelerates the creation process without compromising security level of BRP. Consequently, our system primarily employs pre-key submission mechanism to eliminate potential delays associated with on-demand key submission.

In both mechanisms, relayers submit their public keys to Registration pool on Bifrost Network. Upon completion of key submission, generation of vault on Bitcoin is initiated. After that, initiated address is also accessible through Registration pool on Bifrost Network.

3.1.3 Inbound/Outbound Handler. As described in 3.1, relayers are tasked with identifying BRP-related events and understanding their semantics, such as vault deposits (inbound) and withdrawals (outbound). Inbound handler is responsible for detecting vault deposit events, extracting relevant information, and encapsulating this data into a BRP transaction, which is then forwarded to Transaction manager.

Outbound handler deals with vault withdrawal events by identifying these on Bifrost Network, extracting essential details, and packaging them into a BRP transaction for submission to Transaction manager. This process ensures systematic handling of both inbound and outbound events, maintaining consistency across system's operations.

3.1.4 Transaction Manager. Transaction manager is designated to transmit transactions to Bifrost Network and must accommodate two distinct types of transactions: 1) Extrinsic transactions, which are substrate-level transactions, and 2) EVM transactions, which pertain to Ethereum-level operations. Given that Bifrost Node operates on Substrate frameworks, it is imperative for Transaction manager to adeptly manage both transaction types.

Within context of BRP, Extrinsic transactions facilitate vault generation and PSBT signing for BTC withdrawal. Conversely, EVM transactions are employed for interactions with BRP smart contract on Bifrost Network, such as minting assets on Bitcoin based on deposited BTC. This dual transaction handling capability ensures efficient and precise execution of our protocol operations across different blockchain levels.

3.2 Components within Bifrost Node

In this section, we outline components within Bifrost Node essential for operating BRP. To ensure robust security, BRP utilizes a quorum-based consensus mechanism within its smart contracts on EVM layer, where complex operations such as quorum voting are executed. Additionally, Bifrost Node, built atop Substrate, leverages Substrate-level processing through use of pallets. This approach utilizes two layers for task management. At

Substrate level, it efficiently handles simpler tasks, such as vault management, which complements more complex tasks managed at EVM level. Therefore, this structure boosts processing efficiency while upholding security standards, taking advantage of features like unsigned transactions that don't incur fees.

3.2.1 BRP Smart Contract. EVM Transaction handler interacts with this component that is responsible for message verification and recording entire status of a BRP request. It uses quorum-based consensus mechanism to verify integrity of relayed events. Majority of relayers are required to agree on integrity of relayed events before BRP smart contract will execute corresponding logic (e.g., minting assets on Bifrost Network equivalent to amount of BTC deposited). Components basically use *'transaction id'*, *'address'*, and *'amount'* to verify integrity of relayed events. Furthermore, it emits an EVM event during processing of a BRP request, which includes verifying events occurring on Bitcoin.

3.2.2 BRP Pallets. We have developed specialized pallets tailored for vault generation on Bitcoin and for managing PSBTs. Unlike EVM transactions, BRP Extrinsic transactions do not incur transaction fees, enhancing system's efficiency. First pallet, Socket queue, enables relayers to access unsigned PSBTs. After retrieval, relayers sign these PSBTs with their private keys before returning signed versions to Socket queue for subsequent processing.

Next pallet, Registration Pool, maintains critical information necessary for vault generation, including *'vault address'*, *'signers public key list'*, *'refund address'*, and *'user's EVM address'*. *Vault address* refers to vault's location on Bitcoin. *Signers public key list* encompasses public keys of relayers accountable for vault. *Refund address* is designated to receive BTC in event of a transaction failure, such as when a user sends an amount of BTC below required minimum. Moreover, the address is used in recipient in outbound bridge. *User's EVM address* is Bifrost Network address where user will receive minted Unified BTC (ERC-20) on Bifrost Network. These components collectively ensure seamless operation and security of transactions within BRP framework.

Relayer membership is responsible for verifying authenticity of requests sent by relayers within BRP. This verification process is crucial for both PSBT manager and BRP smart contract to ensure that only legitimate relayers can initiate transactions.

3.3 PSBT Manager

PSBT manager oversees all aspects of PSBT processing, encompassing PSBT broadcasting, initialization, and monitoring of vault generation through BRP smart contract. This is achieved through integration of four sub-modules: Extrinsic handler, Socket listener, PSBT broadcaster, and Vault importer. In the following sections, we will discuss detailed design of each sub-module.

3.3.1 Extrinsic Handler. PSBT manager requires communication with BRP pallets on Bifrost Network to facilitate exchange of PSBTs effectively. To achieve this, Extrinsic handler is essential. It ensures smooth data transfer between PSBT manager and relayer, utilizing Registration Pool and Socket queue for efficient operations. This highly decentralized structure ensures that BRP maintains robust security and reliability standards.

3.3.2 Socket Listener. Socket listener pursues BRP-related events from BRP smart contract. For example, when a user wants to withdraw assets from Bifrost Network, BRP smart contract emits an event to Socket listener. It captures this event and sends corresponding PSBT (withdrawal BTC from Bifrost Vault) to PSBT broadcaster.

3.3.3 PSBT Broadcaster. PSBT broadcaster generates PSBTs in response to BRP-related events. To detect these events, it periodically queries Socket listener—for instance, with each new Bitcoin block—to check if any BRP-related events have occurred. If such events are confirmed, PSBT broadcaster creates PSBTs and forwards them to Socket queue through Extrinsic handler. Additionally, it handles transmission of finalized PSBTs to Bitcoin core, enabling withdrawal of locked BTC to intended recipients.

3.3.4 Vault Importer. Vault importer is designed to ensure that every vault address generated by BRP is recognized and managed by our system. It periodically retrieves list of vault addresses from Registration Pool on Bifrost Network and compares this list with vault addresses currently imported in Bitcoin core. If it detects any new vault addresses, Vault importer requests Bitcoin core to import these addresses, enabling Bitcoin core to manage and facilitate BTC withdrawals from all registered vaults.

In conclusion, PSBT manager is responsible for seamless transfer of assets from vaults, underpinning integrity and functionality of entire BRP framework. Moreover, if PSBT manager is compromised, an attacker could manipulate vault address and relay addresses. However, those malicious actions do not affect user's locked assets; the system might only be paused.

3.4 Bitcoin Core

Bitcoin core serves a crucial function within BRP, as it is responsible for managing and executing transactions involving assets secured within BRP vaults. This component tracks Unspent Transaction Outputs (UTXOs) directed to vault addresses created by BRP system, particularly those listed in Registration Pool, managed by Vault importer. These UTXOs are essential in construction of Partially Signed Bitcoin Transactions (PSBTs), which are prepared and signed by relayers before reaching PSBT broadcaster. When a transaction, such as a BTC withdrawal, is finalized, PSBT broadcaster forwards completed PSBT to Bitcoin core. It then processes transaction, ensuring that BTC is sent to intended recipient, thereby enabling secure transfer of assets from vault. This setup, with exclusive interactions between PSBT manager and this component, reinforces security and reliability of entire BRP framework. Additionally, exclusive access of PSBT manager to Bitcoin core ensures system is sufficiently secure against unauthorized interventions.

3.5 Watchtower: Threat Monitoring System

Watchtower in BRP functions similarly to Risk Management Network of CCIP, actively monitoring and responding to potential threats. It oversees critical aspects of ecosystem, including delays in Web3's RPC servers, unusual changes in BRP smart contracts, and significant shifts in Bitcoin Vault quantities. This comprehensive monitoring ensures system's integrity and smooth operation. Equipped with a Domain-Specific Language (DSL), the component enhances operational efficiency and effectively integrates surveillance rules. By using DSL, it can automatically respond to anomalies based on predefined criteria, such as pausing the bridge, enabling quick adaptation to new threats without extensive manual intervention. Additionally, it is designed to operate universally across all Web3 systems, increasing its utility. We plan to open-source Watchtower to improve security across Web3 platforms like DeFi and NFT marketplaces, providing a robust tool that monitors everything from server performance to asset volumes, thus fostering adoption of proactive security protocols.

3.6 Key Operational Flows

This section outlines primary operational flows, providing a comprehensive understanding of how BRP components interact. Starting with Vault Generation, the process enables asset transfers from Bitcoin to Bifrost Network via Inbound bridge and in reverse via outbound bridge. Operational flows for each module involved in these transactions are described sequentially.

3.6.1 Vault Generation. When a user requests vault generation on Bifrost Network (①), relayers promptly submit their public keys to Registration Pool, which generates a vault address on Bitcoin (②). This address, classified as Pay-to-Witness-Script-Hash (P2WSH), requires a predefined quorum of m out of n relayer signatures to validate transactions, where m is the minimum number of agreeing relayers and n is the total number of executive relayer members. The specific value of m is defined within Bifrost Network to ensure transaction security and consensus. After the vault address is generated by Registration Pool, Vault importer detects it and interfaces with Bitcoin

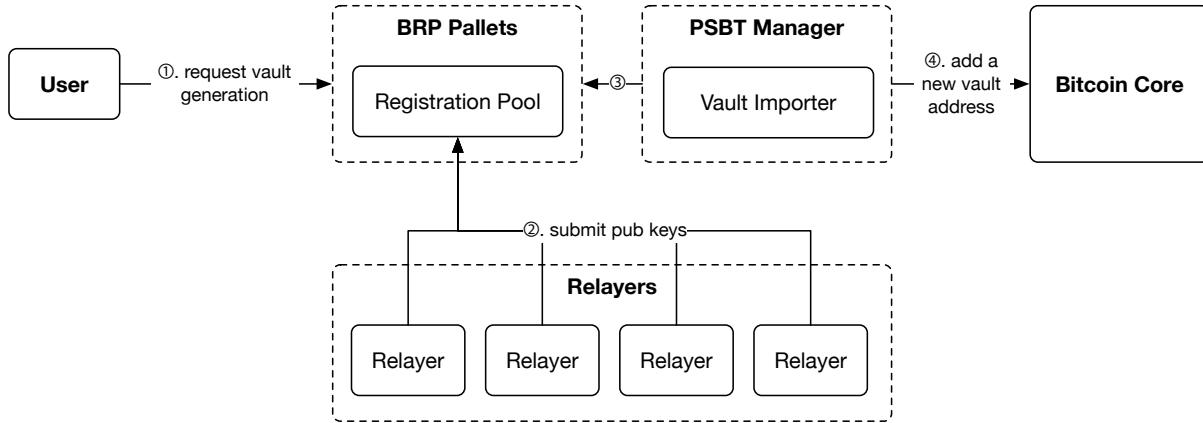


Fig. 4. Vault Generation Process of BRP.

core to import the address (③, ④), enabling management of incoming Unspent Transaction Outputs (UTXOs) directed to the vault (3.4).

All actions, from vault address creation to PSBT management, are meticulously logged within BRP pallets. Users can monitor these processes through a dedicated Bifrost precompile contract for pallets, promoting transparency and trust in the system. This approach ensures secure and verifiable transactions, preserving integrity of operations within BRP framework.

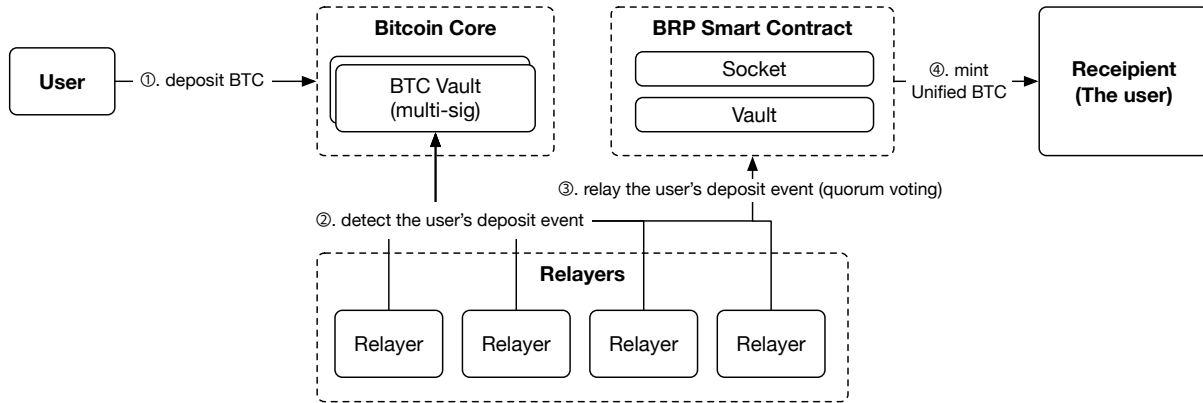


Fig. 5. Inbound Bridge Process of BRP.

3.6.2 Inbound Bridge. Inbound bridge process transfers assets from Bitcoin to Bifrost Network, as depicted in Figure 5. Initially, a user sends x BTC to a BRP vault (①). Relayers then detect this transaction and encapsulate it into BRP transactions (②). Using a quorum-based consensus mechanism, BRP smart contract on Bifrost Network verifies integrity of relayed events (③). Upon successful verification, BRP smart contract mints x Unified BTC (ERC-20) on Bifrost Network, equivalent to deposited BTC (④). Currently, BRP Inbound bridge fees remain free, though this may change in the future.

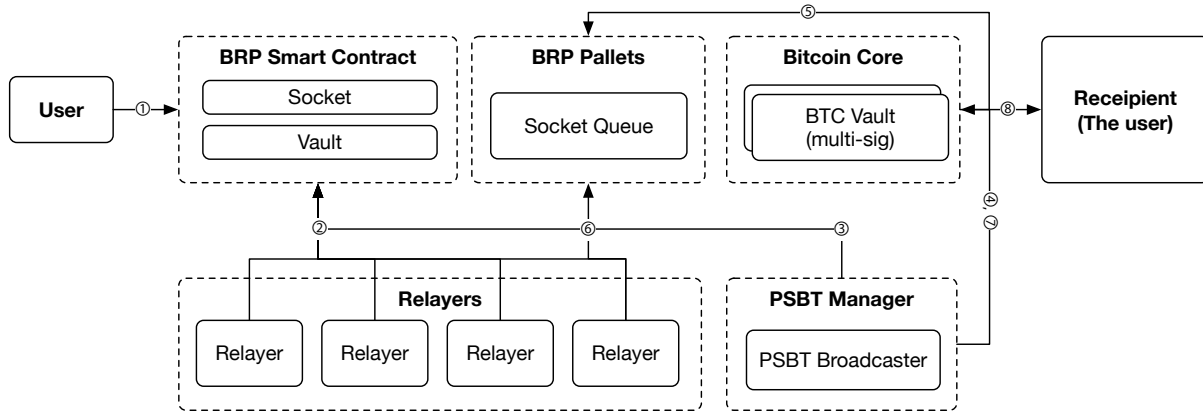


Fig. 6. Outbound Bridge Process of BRP.

3.6.3 Outbound Bridge. Outbound bridge in BRP enables asset transfers from Bifrost Network to Bitcoin. As shown in Figure 6, a user on Bifrost Network deposits Unified BTC into BRP vault (①). relayers then verify deposit transaction and achieve consensus (②). Following verification and consensus, detected by PSBT manager (③), it constructs a PSBT using UTXOs indexed by Bitcoin core (④), which PSBT broadcaster then registers in Socket queue (⑤). relayers subsequently sign PSBT and return signed versions to Socket queue (⑥). Lastly, PSBT manager sends fully signed PSBTs to Bitcoin core (⑦), which then completes withdrawal of locked BTC to intended recipients (⑧).

4 Comparative Advantages of BRP

This section outlines the comparative strengths of BRP over other blockchain technologies. First of all, we examine key aspects where BRP aligns with modern standards, including token compatibility, support for smart contracts, EVM interoperability through Solidity, and a highly energy-efficient Proof of Stake (PoS) consensus mechanism, which together underscore the protocol’s robust security and operational flexibility.

Building on this foundation, we explore the unique advantages that distinguish BRP. Enhanced interoperability enables BRP to seamlessly integrate across multiple blockchain networks, expanding Bitcoin’s functionality beyond traditional transactions. As a fully accountable protocol, BRP meticulously records all activities, ensuring transparency and verifiability at every step. The federated relayer network further enhances security and decentralization by engaging diverse participants in network operations. Additionally, the omnichain framework simplifies cross-chain asset management, easing liquidity handling for DApp developers. These combined features establish BRP as a scalable, forward-looking solution, advancing Bitcoin’s role within an interconnected blockchain ecosystem.

4.1 Comparison with Other Protocols

We provide a detailed analysis of how BRP stacks up against various other blockchain platforms based on several key technological attributes. This comparative overview highlights BRP’s distinct advantages, demonstrating its robust capabilities and areas where it aligns closely with industry standards.

Token Compatibility and Smart Contracts: Unlike Bitcoin and other protocols such as Lightning and Merlin that do not support smart contracts, BRP enables the use of decentralized applications (DApps) by supporting

	Bitcoin	<i>BRP</i>	Lightning	Stacks	Rootstock	Liquid	Merlin
TK	BTC	Unified BTC	BTC	sBTC	RBTC	L-BTC	MBTC
SC	✗	✓	✗	✓	✓	✓	✗
PL	Bitcoin Script	Solidity	-	Clarity	Solidity	Simplicity	-
EC	✗	✓	✗	✗	✓	✗	✗
CM	PoW	PoS	-	PoX	PoW	Federated Sidechain	-

Table 1. Comparison of Blockchain Platforms. TK stands for Token, SC stands for Smart Contract, PL stands for Programming Language, EC stands for EVM Compatible, and CM stands for Consensus Mechanism.

smart contracts. This is akin to platforms like Stacks, Rootstock, and Liquid, which also facilitate smart contracts, indicating BRP’s alignment with more dynamic aspects of blockchain functionalities.

Programming Language and EVM Compatibility: BRP’s use of Solidity, a widely adopted programming language associated with Ethereum, sets it apart from Bitcoin and other systems that rely on less common languages like Bitcoin Script, Go, or Clarity. Supporting EVM compatibility, this choice is crucial for developers aiming to create versatile and interoperable applications. By allowing developers to work in a familiar language, BRP minimizes the learning curve. Similar to Rootstock, it provides a bridge for developers accustomed to the Ethereum environment to engage with Bitcoin, enhancing the developer experience without sacrificing functionality.

Consensus Mechanism: BRP operates on a Proof of Stake (PoS) consensus mechanism, celebrated for its efficiency and significantly lower energy consumption compared to the Proof of Work (PoW) used by Rootstock. This makes it a more sustainable alternative, aligning with the ongoing shift towards environmentally friendly blockchain solutions. Unlike Stacks, which uses the PoX (Proof of Transfer) protocol, PoS does not rely on continuous Bitcoin transactions to secure the network, thereby reducing dependency on external blockchains and avoiding PoX’s challenges with high volatility and operational complexity. In contrast to BRP and other permissionless protocols, Merlin and Lightning employ centralized architectures, sacrificing decentralization and open access for speed—thus potentially limiting scalability and increasing reliance on trusted entities within the network.

In conclusion, BRP not only matches up well against other protocols in fundamental aspects such as security, smart contract capability, and programming flexibility but also excels in providing a gateway to expansive blockchain interoperability and sustainable blockchain practices. This comprehensive approach ensures BRP’s position as a crucial tool in the broader adoption and utility of Bitcoin within the ever-evolving blockchain landscape.

4.2 Fully Accountable Protocol

The outstanding feature of BRP is its full accountability. To achieve this, BRP records all activities on the Bifrost Network, capturing relayer signatures, transaction details, and every critical step of the bridging process. This comprehensive record-keeping ensures that all activities are transparent and fully verifiable, establishing a high level of accountability for all participants. By enabling real-time visibility into each transaction and relayer action, BRP stands out as a uniquely transparent and accountable bridge protocol. This level of verifiability not only fosters trust among users but also positions BRP as a leading solution in the market for secure and transparent cross-chain bridging.

4.3 Strong Federated Relay Network

Bifrost Network embodies a fully decentralized model, inviting a broad spectrum of participants to become node operators. This inclusivity not only extends to block validators but also to relayers, thereby enhancing the network's robustness and resilience. Currently, our relay network boasts a diverse and powerful coalition of stakeholders, including venture capitalists, cryptocurrency exchanges, blockchain security firms, and technically proficient individuals. This eclectic mix ensures a rich, well-rounded perspective on network operations and continuous innovation.

To maintain operational excellence and security, we facilitate direct, ongoing communication with all operators through a dedicated hotline. This approach has proven effective, as demonstrated by our excellent security record with no incidents reported over the past two years. This track record highlights the reliability of our network and showcases the strong federated structure of the relay network, making it a cornerstone of the success of Bifrost Network in enabling seamless cross-chain transactions.

4.4 Omnichain Ecosystem

Bifrost Network stands out as a robust omnichain platform, uniquely positioned to serve as an intersection for both EVM and non-EVM blockchain networks, unlike other Bitcoin protocols (as shown in Table 1) that are typically confined to single-chain operations. This omnichain interoperability allows users to enjoy seamless interactions without the need to deploy assets on any specific blockchain. Such capability significantly eases the management of fragmented liquidity and reduces operational costs for DApp developers, enhancing convenience and efficiency in deploying and managing decentralized applications across various chains. Currently, our ecosystem supports a diverse range of chains including Bitcoin, Ethereum, Binance Smart Chain, Polygon, Arbitrum, and Base, with plans to extend interoperability to even more blockchain networks. This strategic expansion not only extends Bifrost's utility but also establishes it as a key contributor to building a more interconnected and inclusive blockchain ecosystem.

5 Conclusion

The BRP, based on the well-established CCCP, marks a strategic advance in Bitcoin's functionality, integrating it more deeply within the broader blockchain ecosystem. This development is pivotal as it transforms Bitcoin from a mere store of value into a dynamic platform capable of supporting diverse decentralized applications and systems across both non-EVM and EVM blockchains.

As we continue to enhance BRP, our vision is clear: to bridge Bitcoin with multiple blockchain technologies seamlessly, making it a cornerstone for innovation and interoperability. This effort not only underscores our commitment to pushing the boundaries of blockchain technology but also positions BRP as a critical tool in shaping the future of decentralized networks. By leveraging the robust security and proven efficiency of CCCP, BRP is set to redefine blockchain interoperability and expand Bitcoin's role within the global tech landscape.